An Adaptive Blind Video Watermarking Technique based on SD-BPSO and DWT-SVD

Prathik P
Student
Dept. Electronics and Comm.
M.S. Ramaiah Inst. Of Tech.
Bangalore 560054, India
prathik37@gmail.com

Rahul Krishna
Student
Dept. Electronics and Comm.
M.S. Ramaiah Inst. Of Tech.
Bangalore 560054, India
i.m.ralk@gmail.com

Rahul Ajay Nafde
Student
Dept. Electronics and Comm.
M.S. Ramaiah Inst. Of Tech.
Bangalore 560054, India
rahulanafde@gmail.com

Shreedarshan K*
Asst. Professor
Dept. Electronics and Comm.
M.S. Ramaiah Inst. Of Tech.
Bangalore 560054, India
shreedarshan.k@msrit.edu

*Corresponding Author

Abstract— **Data Exchange in the form of multimedia has seen a tremendous increase in the past decade thereby requiring a need for better security and protection for proprietary rights. Blind Watermarking is a well-established authentication technique and in this paper, we propose a novel algorithm for watermarking videos. The proposed concepts include *Adaptive Frame Selection* using SD-BPSO to ensure the watermarks have least detrimental effects on the video as a whole. The integrity of the video is validated using the Peak Signal to Nose Ratio (PSNR). The robustness of the algorithm is also tested by subjecting the videos to several standard attacks such as -- rotation, cropping, image shift, histogram equalization and image sharpening. Bit Error Rate (BER) is also used, in order to determine the efficiency of the system in retaining the watermark.**

Keywords- Video watermarking, Blind watermarking, Binary Particle Swarm Optimization (BPSO), Standard Deviation, Discrete Wavelet Transform, Singular Value Decomposition.

I. INTRODUCTION

Watermarking is a branch of information hiding which is used to hide proprietary information in digital media. [1], [2], [3], [4], [5] provides excellent insight into watermarking. There has been mounting interest with reference to digital watermarking all through the past decade certainly due to the increase in the need for copyright protection. Applications of video watermarking are multifold— in copy control, broadcast monitoring, video authentication and copyright protection to name a few. There has also been a marked rise in copyright infringement due to the relative ease with which digital content can be exchanged over peer-to-peer network. In order to protect the interest of the content providers, these digital contents must be watermarked. In this paper the authors have designed a blind video watermarking technique based on a new variant of the Binary Particle Swarm Optimization called the SD-BPSO. The watermark thus embedded cannot be easily removed, without a significant degradation of the video sequence, from the watermarked signal even after being subjected to a number attacks, both intentional and unintentional.

The rest of the paper is organized as follows— the subsequent section discusses the preliminaries and related work, section III highlights the proposed algorithm and presents a detailed analysis of the same. Section IV validates the performance of the proposed algorithm. Section V concludes our work with a comprehensive summary.

II. PRELIMINARIES AND RELATED WORK

A. YCbCr Color System

One of the alternative forms of representation of an image is the YCbCr color system which is used in video and digital photography systems. In this system, the image is split into three components namely, Y (luma component), Cb (blue difference) and Cr (red difference). Usually, images are represented in the RGB form; however, a translation into this system proves to be more efficient as the native (RGB) format has a lot of redundant values. The conversion form RGB format to YCbCr format is given by Eqs. (1–3).

$$Y=0.299R+0.587G+0.114B \quad (1)$$

$$Cr=0.5R+0.41869G-0.08131B+128 \quad (2)$$

$$Cb=-0.16874R-0.33126G+0.5B+128 \quad (3)$$

B. Binary Particle Swarm Optimization (BPSO)

PSO is an evolutionary optimization algorithm based on swarm behavior proposed by [6]. The algorithm introduces the concept of particles, each which represent a candidate solution. The algorithm is modeled by taking into account the social and cognitive influence factors inherent in swarm behavior. The first step involves initialization of possible paths defined by the particle size, to the goal state. The algorithm seeks to converge to the optimum path by using a heuristic defined by the fitness function. In this paper a discrete binary version of the particle swarm optimization method is used which was proposed by [7]. In the discrete version of Particle Swarm Optimization (PSO) each particle's position is a string of 1's or 0's. The positional values are determined by the sigmoid function and probabilistic rule. The particle velocity is modeled as a probabilistic function for positional update. Potential solution is represented as a particle having positional co-ordinates $X^t_i = [x_{i1}, x_{i2},\ldots x_{iD}]$ in a D dimensional space where i denotes the particle number and 't' represents the iteration number. Each particle maintains a record of the position of its previous best performance in a personal best position vector P_{best}. An iteration comprises evaluation of each

978-1-4673-2907-1/13/$31.00 ©2013 IEEE

particle and then stochastic adjustment of its velocity $V_i^t = [v_{i1}, v_{i2},\ldots,v_{iD}]$ in the direction of its own previous best and the best previous position of any particle in the neighborhood. The best position of any individual in the whole swarm is stored as the global best position G_{best}. PSO is described by the following velocity and position update equations:

$$V_i^{t+1} = w.V_i^t + c_1.rand.(Pbest_i - X_i^t) + c_2.rand.(Gbest - X_i^t) \quad (4)$$

Where w = inertia weight, c_1 = cognitive parameter, c_2 = social parameter.

$$X_i^{t+1} = X_i^t + V_i^{t+1} \quad (5)$$

For i=1 to P; P = number of particles. If r is a random number between 0 and 1, the equation that updates the particle position is:

$$X_i^{t+1} = 1 \text{ if } r < \frac{1}{1+e^{-V_i^{t+1}}}, \text{ else } X_i^{t+1} = 0. \quad (6)$$

C. Standard Deviation.

It is a mathematical tool which is used to indicate variation or dispersion of values from the average (mean, or expected value) from a given set of values given by Eq. 7. A low standard deviation indicates that the data points tend to be very close to the mean, whereas high standard deviation indicates that the data points are spread out over a large range of values.

$$\sigma = \sqrt{\frac{1}{M}[(y_1-\mu)^2 + (y_2-\mu)^2 + \ldots + (y_M-\mu)^2]} \quad (7)$$

Where σ = standard deviation, μ = mean; $y_1, y_2, \ldots, y_N$ are the set of all M values. And,

$$\mu = \left(\frac{1}{M}\right) \times \sum_{k=1}^{M} y_k$$

D. Discrete Wavelet Transform (DWT)

The DWT (Discrete Wavelet Transform) is a spatial transform that separates an image into a lower resolution approximation image (LL) as well as horizontal (HL), vertical (LH) and diagonal (HH) detail components. The process can then be repeated to compute multiple scale wavelet decomposition using approximation image (LL). Embedding watermarks in LH, HL and HH regions allow us to increase the robustness of our watermark, at little to no additional impact on image quality. DWT-based watermarking scheme is the most robust to noise addition.

For 2-D images, applying DWT corresponds to processing the image by 2-D filters in each dimension. The filters divide the input image into four non-overlapping multi-resolution sub-bands LL, LH, HL and HH. Fig. 1 shows these four components of an image. The LL sub-band represents the coarse-scale DWT coefficients while the LH, HL and HH sub-bands represent the fine-scale DWT coefficients.

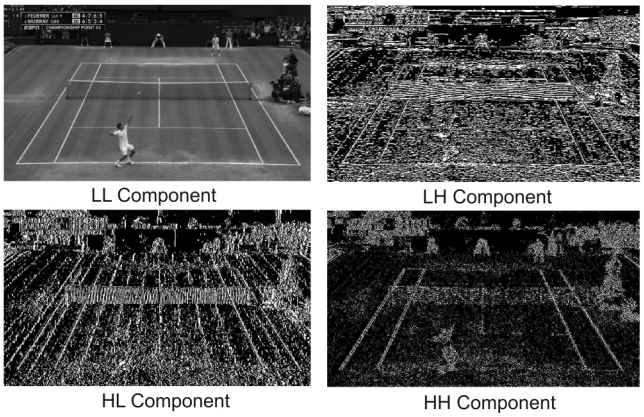


Fig. 1. DWT coefficients

E. Singular Value Decomposition (SVD)

Singular value decomposition (SVD) is a numerical technique used to diagonalize matrices in numerical analysis. It is an algorithm developed for a variety of applications. The main properties of SVD are:

- The singular values (SVs) of an image have very good stability, i.e., when a small perturbation is added to an image, its SVs do not change significantly.
- SVs represent intrinsic algebraic image properties.

III. PROPOSED ALGORITHM

The generic watermarking system consists of two phases. The first being the watermarking phase shown in Fig. 2 followed by the extraction of watermark and authentication shown in Fig. 3. Principal contributions have been made in the watermarking stage and the following section provides a formal discourse of the same.

A. SD-BPSO based frame Isolation:

A new variant of BPSO called the SD-BPSO is proposed here. The SD-BPSO algorithm is a novel technique which selects potential frames from the original images where the watermark may be embedded in order to achieve maximum PSNR. It also generates a unique key which can then be used to extract and evaluate the watermark.

A flow chart of the SD-BPSO is shown in Fig. 4. To begin with, each video sequence is split into its corresponding frames. This allows the SD-BPSO algorithm to process the video as individual images whilst retaining the correspondence between the subsequent frames.

The SD-BPSO algorithm begins with ‘Q’ particles each positing potential frames to be watermarked. A particle is a binary array of size 1xN where N is the number of frames in the original video sequence. In this array, a binary ‘1’ implies that the frame is to be watermarked and binary ‘0’ skips the frame. For example, a 24fps video lasting for 20 seconds would ideally contain 480 frames; each particle in BPSO will then

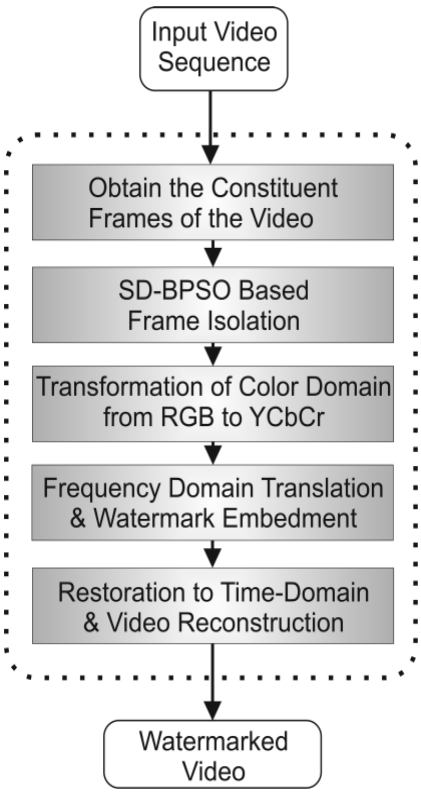


Fig. 2. Top Level Overview of the proposed watermarking Stage

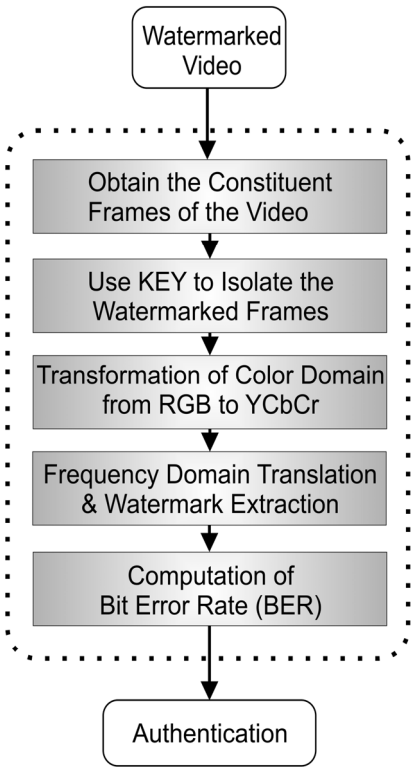


Fig. 3. Top level Overview of the corresponding Extraction Stage

have a size of 1480 which may resemble Fig. 5. Initial frame estimate of each of the particle is purely random. A measure to evaluate the solutions of each of the particle is required. We propose the use of standard deviation Eq. 7 as a fitness evaluator, playing a crucial role in the algorithm. The Standard Deviation is obtained for every frame of the video sequence. The net standard deviation is computed using Eq. 8 by adding the standard deviations of only those frames that were chosen by a particle.

$$SD_{net} = \sum_{i=1}^{N} P_i \times SD(F_i) \tag{8}$$

Where, N is the total number of frames, SD(•) is the standard deviation, F_i is the ith frame, and P_i is the particle's value corresponding to the ith frame, which can be written as

$$P_i = \begin{cases} 1 & \text{if frame } F_i \text{ is to be watermarked} \\ 0 & \text{if frame } F_i \text{ is to be skipped} \end{cases}$$

This value is then normalized over the total standard deviation of all frames inclusive. This will be the fitness function and it can be mathematically expressed as in Eq. 9

$$\text{Fitness} = \frac{SD_{net}}{SD_{total}} \tag{9}$$

Where, SD_{total} is the sum of standard deviations of all frames. This process is repeated for frames suggested by each of the 'Q' particles. At the end of the BPSO algorithm, the particle with the highest fitness value is chosen and those frames are watermarked. Best particle's array can also be used as the 'KEY' which will then be used by the receiver in order to extract the watermark.

B. Conversion to YCbCr color domain

The frames that are to be embedded are selected using the SD-BPSO technique, the selected frames in RGB color domain are converted to YCbCr domain, then Y (luma) component is further used in embedding watermark. RGB color domain is not used directly since any changes in this system will lead to loss of imperceptibility.

C. Frequency Domain Translation and Watermarking

DWT technique is applied to the luma component from the selected frame, leading to 4 bands, LL,LH,HL,HH as explained previously in Section II-D, this was used in [8], [9]. In our technique we convert the watermark into a 1D array Fig. 6 and embed pre-determined number of bits of watermark into every frame till all the bits of the watermark are exhausted. The low frequency band (LL) is not used because any changes in this band will be reflected directly. Thus other 3 bands are used for watermarking. We embed a third of the bits into LH, HL and HH bands each of every frame. We use a technique called Singular Value Decomposition to put in these bits.

Embedding Process

The first step in embedding is the application of SVD transform to each wavelet (except to the LL wavelet) individually. This produces 3 matrices U, S and

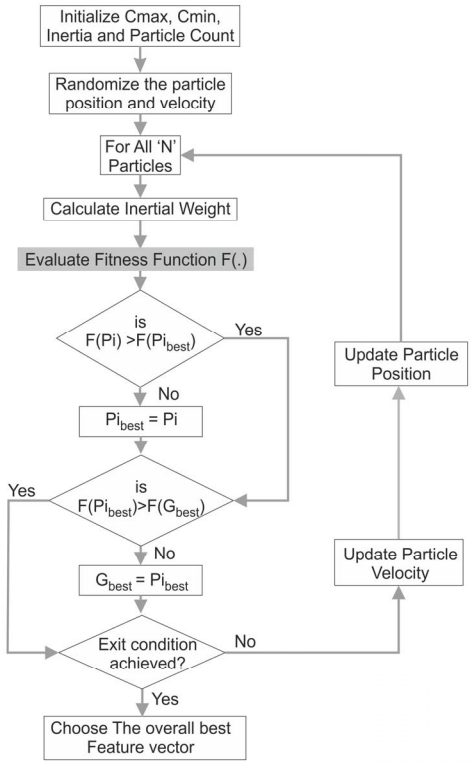


Fig. 4. Flowchart of the proposed SD-BPSO Algorithm

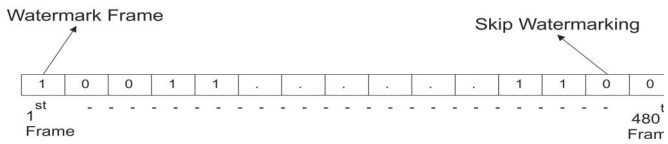


Fig. 5. The constituents of a particle for a 24fps video lasting 20 seconds.

V. Then if A were to be the wavelet, we may express it as

$$A = USV^T \tag{10}$$

Here, if s_1 and s_2 are the singular values present in the S matrix then columns of U and V are, respectively, left and right singular vectors for the corresponding singular values. The singular values in S matrix are modified to maintain the visibility and robustness of wavelet and among the S matrix only middle singular values are chosen. The watermarking of each bit is modeled using the following equation

$$s'_i = \frac{1}{2}\left(\left(s_{i-1}+s_{i+1}\right)+\alpha\times w_i\times\left(s_{i-1}+s_{i+1}\right)\right) \tag{11}$$

Here, S_i is individual element in the S matrix; α is the watermark intensity and w_i is the bit to be embedded. The news matrix after embedding may be denoted as S'.

Once all the bits have been embedded, inverse SVD is then used to obtain the modified wavelets. The inverse SVD is given by

$$A'=US'V^T \tag{12}$$

$$S'=\mathrm{diag}(s'_1,s'_2,\ldots,s'_r) \tag{13}$$

After the modified wavelets are obtained Inverse DWT is applied, the image thus obtained is the modified luma component of the original image. This is then concatenated with the unchanged Cb and Cr components to obtain the watermarked image in YCbCr color domain; this is converted to RGB domain and is now watermarked.

This process is repeated for all the selected frames until the bits of watermark have been successfully embedded. After all frames have been watermarked, the video sequence is reconstructed to obtain the watermarked video.

D. Extraction

In the extraction process the watermarked video is broken down into frames. The only parameter that is required at the receiver end is the content of the best particle's array (the KEY). This is used to select the watermarked frames among all the frames that are present. The selected frames are in RGB domain; these are converted to YCbCr color domain and only the Y component is considered because the watermarked bits are embedded in component. DWT transform is applied to the luma component; this gives rise to four sub-bands. Except the LL component, the other wavelet components are considered. Let one of the sub-band be denoted as A'. Then the SVD transform is applied to obtain the modified matrix S'. This modified matrix contains the bits that are embedded. These bits are extracted using Eq. 14

$$w_i = \begin{cases} 0 \text{ if } s'_i > 0.5\times\left(s'_{i-1}+s'_{i+1}\right) \\ 1 \text{ if } s'_i < 0.5\times\left(s'_{i-1}+s'_{i+1}\right) \end{cases} \tag{14}$$

This is repeated for all sub-bands in the watermarked frames. It is to be noted that the extraction process from the wavelets must follow the exact same order as prescribed above.

Similar methods have also been found in [10], [11], [12], [13]. The watermark thus extracted from the video sequence is strikingly similar to the watermark used for the embedding process albeit with subtle variations, refer Fig. [6].

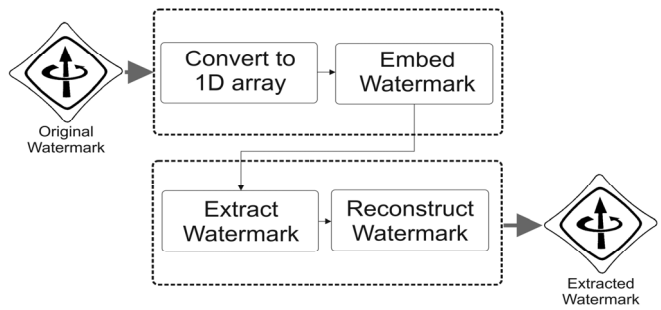
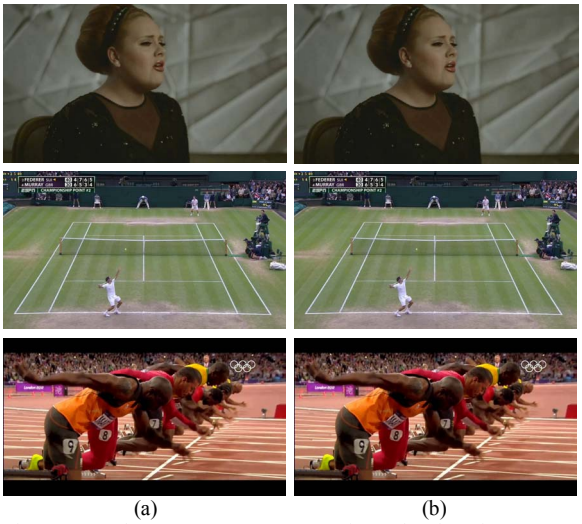


Fig. 6. Watermarking and Extraction

IV. EXPERIMENTAL RESULTS

To test the algorithm, videos were chosen randomly from the internet [18], [19], [20]. Frames of these videos are shown in Fig. 7a. The watermark shown in Fig. 6 was resized and converted into an 1D array using raster scanning technique, and

the resulting vector was embedded in all the three videos. All the video sequences are of the resolution 1280 x 768 and the length of the watermark sequence is 1380 bits. All the videos were watermarked using the technique we have proposed in the aforementioned sections. Screenshots of the frames in Fig. 7a are watermarked and are shown in Fig. 7b. All the codes were implemented on MATLAB [17].



(a) (b)

Fig. 7. Top to bottom– "Woman", "Tennis" and "Olympics".
a) Original frames. b) Watermarked frames.

To validate the robustness of the algorithm, the Peak Signal to Noise Ratio (PSNR) and Bit Error Rate (BER) are computed. PSNR depends on Mean Squared Error (MSE), given by Eq. 15

$$\mathrm{MSE} = \frac{\sum_{M,N}\left[I_1(m,n)-I_2(m,n)\right]^2}{M\times N} \tag{15}$$

Here, M and N are rows and columns of an image. With this PSNR can be calculated using Eq. 16

$$\mathrm{PSNR}=10\log_{10}\left(\frac{R^2}{\mathrm{MSE}}\right) \tag{16}$$

R is the maximum fluctuation in the input image data. BER is the Bit Error Rate and it can be defined as the number of bits that are erroneous in the extracted watermark to the total number of bits in the original watermark.

TABLE I.
PSNR OF THE FIRST 30 SELECTED FRAMES OF "WOMEN" VIDEO

Frame Number	**PSNR (dB)**				
1 to 5	86.39	80.68	84.19	88.16	86.46
6 to 10	86.06	75.23	90.00	89.21	80.09
11 to 15	84.30	83.15	82.21	85.79	87.82
16 to 20	80.03	84.74	81.99	86.28	92.25
21 to 25	90.02	88.49	89.09	86.98	87.09
26 to 30	91.13	70.61	78.29	89.60	86.58

PSNR of the first 30 selected frames of "Women" video is shown in Table I. PSNR for 100 frames of all the three videos are plotted in a graph as shown in Fig. 8.

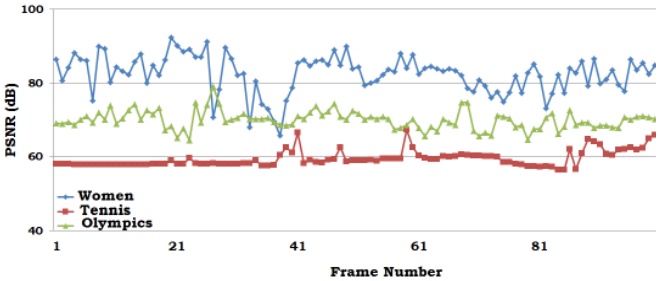


Fig. 8. PSNR variation with frame number for selected video samples

(a) Original Image (b) Rotation

(c) Crop (d) Shift

(d) Median Filter (e) Sharpen

Fig. 9. Attacks on watermarking video.

Video sequences may also undergo a lot of changes due to various factors. This effect is called an 'attack', these attacks may either be intentional or may be unintentional, but for a watermark to be robust it should be resistant to any such attack. Different attacks are considered and their effect on a particular video is shown in Fig. 9. The efficacy of the algorithm to withstand these attacks is evident in the results tabulated in Table II. To obtain the results that were tabulated in Table II, all the frames were exposed to the attack. The BER thus calculated is the absolute maximum error rate that can occur under any circumstance. The technique is promising, in that the results obtained are on par with, and in many cases much better than, those presented by the work of [14], [15], [16].

V. CONCLUSIONS

The proposed algorithm is frame adaptive due to the use of swarm optimization for selection of frames, it is also a blind watermarking technique in that a measure (BER) is used to compare and find out the strength of the watermark. The

TABLE II

MAXIMUM BER AND PSNR RANGE FOR DIFFERENT ATTACKS ON THE VIDEO SEQUENCE

Video	Attack	BER	PSNR range (dB)
Women	Image rotation by 5 degrees	0.3872	10 - 20
	Cropping out (100x100) pixels	0.0285	20 – 32
	Image shift by 10 rows and 10 columns	0.0579	12 - 22
	Histogram Equalization to enhance contrast	0.0381	5 -14
	Image Sharpening using unsharp mask	0.1743	23 - 39
Tennis	Image rotation by 5 degrees	0.4625	10 - 15
	Cropping out (100x100) pixels	0.0302	22 - 27
	Image shift by 10 rows and 10 columns	0.0023	12 - 15
	Histogram Equalization to enhance contrast	0.0539	07 – 12
	Image Sharpening using unsharp mask	0.2125	17 – 22
Olympics	Image rotation by 5 degrees	0.4342	10 - 17
	Cropping out (100x100) pixels	0.029	20 - 28
	Image shift by 10 rows and 10 columns	0.0492	14 - 21
	Histogram Equalization to enhance contrast	0.0635	02 - 10
	Image Sharpening using unsharp mask	0.1934	16 - 25

PSNR values obtained as shown in Fig 8. clearly shows that the proposed algorithm will serve as a good watermarking agent. It can be observed that the proposed watermarking algorithm can be extended to larger vector lengths. Table II helps us better understand about the efficacy of the algorithm, it can be seen that the number of bits that are in error are negligible in case of Cropping, Image shift and Histogram equalization. SD-BPSO is a novel algorithm which serves as an effective optimizer by selecting suitable frames for better imperceptibility.

REFERENCES

[1] C. I. Podilchuk and E. J. Delp, Digital Watermarking: Algorithms and Applications, IEEE Signal Processing Magazine, July 2001, pp. 33-46.

[2] E. Vellasques, E. Granger, R. Sabourin, Handbook of Pattern Recognition and Computer Vision, 4th ed., World Scientific Review, 2010, Ch. Intelligent Watermarking Systems: A Survey., pp. 687-724.

[3] I. J. Cox, M. L. Miller, and J. A. Bloom, Digital Watermarking, Morgan Kaufmann Publishers, 2002.

[4] R.G.V. Schyndle, A.Z. Tirkel, C.F. Osbrone, A digital watermark, Proceedings of IEEE International Conference on Image Processing, vol. 2, 1994, pp. 86-90.

[5] Vidyasagar M. Potdar, Song Han, Elizabeth Chang, A Survey of Digital Image Watermarking Techniques, 3rd International Conference on Industrial Informatics(INDIN 2005)

[6] J. Kennedy, R. C. Eberhart, Particle Swarm Optimization, IEEE International Conference on Neural Networks, 1995.

[7] J. Kennedy, R. C. Eberhart, A Discrete Binary Version of the Particle Swarm Algorithm, Proceedings of IEEE International Conference on Systems, Man, and Cybernetics, vol. 5, p. 4104-4108, October 1997.

[8] Kundur. D., Hatzinakos, D., Digital Watermarking using Multiresolution Wavelet Decomposition, Proc. IEEE Int. Conf. On Acoustics, Speech and Signal Processing, Seattle, Washington, vol. 5, pp. 2969-2972, May 1998.

[9] Kaewkamnerd, N., Rao, K.R., Wavelet based image adaptive watermarking scheme in IEE Electronics Letters, vol.36, pp.312-313, 17 Feb.2000

[10] Gaurav Bhatnagar and Balasubramanian Raman, A new robust reference watermarking scheme based on DWT-SVD, Computer Standards and Interfaces 31 (2009), pp.1002-1013

[11] R. Liu, T. Tan, An SVD-based watermarking scheme for protecting rightful ownership, IEEE Transactions on Multimedia 4 (1) (2002), pp.121-128.

[12] V.Santhi and Dr. Arunkumar Thangavelu, DWT-SVD Combined Full Band Robust Watermarking Technique for Color Images in YUV Color Space, International Journal of Computer Theory and Engineering, Vol. 1, No. 4, October2009, pp. 1793-8201

[13] Wenhai Kong, Bian Yang, Di Wu, and Xiamu Niu1, SVD Based Blind Video Watermarking Algorithm, Proceedings of the First International Conference on Innovative Computing, Information and Control, 2006

[14] Qiang Li, et al, Adaptive DWT-SVD Domain Image Watermarking Using Human Visual Model proceedings of 9th international conference on advanced communication Technology, Volume 3, pp. 1947 - 1951, Feb.2007.

[15] Cheng-qun Yin, Li Li, An-qiang Lv and Li Qu, Color Image Watermarking Algorithm Based on DWT-SVD Proceedings of the IEEE International Conference on Automation and Logistics, pp. 2607 2611, August 18 - 21, 2007

[16] Qing-Ming Ge,Zhe-Ming Lu, Xia-Mu Niu. Oblivious Video Watermarking Scheme with Adaptive Embedding Mechanism. Machine Learning and Cybernetics, 2003 International Conference on Volume 5, 2-5 Nov, pp. 2876 - 2881

[17] Matlab, www.mathworks.com/products/matlab/

[18] Olympics, http://www.youtube.com/watch?v=meF60gScJ5g

[19] Women, http://www.youtube.com/watch?v=rYEDA3JcQqw

[20] Tennis, http://www.youtube.com/watch?v=z rJYiTkmu8